

States, Evolution Laws, Flows, First-Order Systems, Vector Fields, and Phase Portraits

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Idea: A dynamical model

A deterministic dynamical model consists of

$$\boxed{\text{state}} \quad + \quad \boxed{\text{evolution law}} \quad + \quad \boxed{\text{initial condition}}.$$

The state records the information needed to apply the evolution law.

Definition: Continuous-time dynamical system

Let $n \geq 1$, let $X \subseteq \mathbb{R}^n$, let $J \subseteq \mathbb{R}$ be an interval, and let $f : J \times X \rightarrow \mathbb{R}^n$. A **continuous-time dynamical system** is an initial-value problem

$$\dot{x} = f(t, x), \quad x(t_0) = x_0, \quad (t_0, x_0) \in J \times X.$$

The set X is the **state space**, also called the **phase space**. The system is **autonomous** if there is a function $g : X \rightarrow \mathbb{R}^n$ such that $f(t, x) = g(x)$ for every $(t, x) \in J \times X$.

Notation: Time derivative

The variable t denotes time, and

$$\dot{x} = \frac{dx}{dt}.$$

Recall: Solution and trajectory

A **solution** of $\dot{x} = f(t, x)$ through (t_0, x_0) is a differentiable function $x : I \rightarrow X$, where $I \subseteq J$ is an interval containing t_0 , such that

$$x(t_0) = x_0, \quad \dot{x}(t) = f(t, x(t)) \quad (t \in I).$$

Its image $\{x(t) : t \in I\}$ is its **trajectory**, or **orbit**.

Definition: Periodic solution

A solution $x : \mathbb{R} \rightarrow X$ is **periodic** if some $T > 0$ satisfies

$$x(t + T) = x(t)$$

for every $t \in \mathbb{R}$. If x is nonconstant, its trajectory is a **periodic orbit**.

Definition: Equilibrium

For an autonomous system $\dot{x} = g(x)$, a point $x^* \in X$ is an **equilibrium**, also called a **fixed point**, if

$$g(x^*) = 0.$$

Observation: Equilibria and constant solutions

For the autonomous system $\dot{x} = g(x)$, the constant function

$$x(t) = x^*$$

is a solution if and only if x^* is an equilibrium.

Definition: Local and global flow

Let $X \subseteq \mathbb{R}^n$ be open, and let $g \in C^1(X, \mathbb{R}^n)$. For each $x_0 \in X$, let I_{x_0} be the largest interval containing 0 on which the unique solution $x(\cdot; x_0)$ of

$$\dot{x} = g(x), \quad x(0) = x_0,$$

is defined. Define

$$\mathcal{D} = \{(t, x_0) : x_0 \in X, t \in I_{x_0}\}.$$

The **local flow** is

$$\Phi : \mathcal{D} \rightarrow X, \quad \Phi(t, x_0) = x(t; x_0).$$

It is a **global flow** if $I_{x_0} = \mathbb{R}$ for every $x_0 \in X$.

Notation: Time-indexed flow map

For each $t \in \mathbb{R}$, define

$$X_t = \{x \in X : (t, x) \in \mathcal{D}\}.$$

The corresponding **time- t map** is

$$\Phi_t : X_t \rightarrow X, \quad \Phi_t(x) = \Phi(t, x).$$

Recall: Action of the real numbers

An **action of** $(\mathbb{R}, +)$ **on** X is a map $A : \mathbb{R} \times X \rightarrow X$ satisfying

$$A(0, x) = x, \quad A(t + s, x) = A(t, A(s, x))$$

for all $s, t \in \mathbb{R}$ and $x \in X$.

Observation: Autonomous flow law

For the local flow of an autonomous system,

$$\Phi(0, x) = x, \quad x \in X.$$

If $x \in X_s$ and $\Phi_s(x) \in X_t$, then

$$x \in X_{t+s}, \quad \Phi_{t+s}(x) = \Phi_t(\Phi_s(x)).$$

For a global flow, $X_t = X$ for every $t \in \mathbb{R}$, so

$$\Phi_0 = \text{id}_X, \quad \Phi_{t+s} = \Phi_t \circ \Phi_s.$$

Thus a global flow is an action of $(\mathbb{R}, +)$ on X .

Recall: Reducing an equation to first order

Let $n \geq 1$ be an integer and let $G : \mathbb{R}^n \rightarrow \mathbb{R}$. For the n th-order scalar equation

$$q^{(n)} = G(q, \dot{q}, \dots, q^{(n-1)}),$$

define

$$x_1 = q, \quad x_2 = \dot{q}, \quad \dots, \quad x_n = q^{(n-1)}.$$

Then

$$\dot{x}_1 = x_2, \quad \dots, \quad \dot{x}_{n-1} = x_n, \quad \dot{x}_n = G(x_1, \dots, x_n).$$

The resulting first-order system has state $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. Its Euclidean phase-space dimension is n .

Question: What is the state of a damped oscillator?

Rewrite

$$m\ddot{q} + b\dot{q} + kq = 0, \quad m > 0, \quad b \geq 0, \quad k > 0,$$

as a first-order autonomous system.

Solution. Define

$$x_1 = q, \quad x_2 = \dot{q}.$$

Then

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2,$$

or

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -k/m & -b/m \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Question: What is the state of a damped oscillator? (continued)

The state is $(q, \dot{q}) \in \mathbb{R}^2$. Position alone is insufficient because the same position can occur with different velocities.

Definition: Vector field

Let $X \subseteq \mathbb{R}^n$ be open. A **vector field** on X is a function

$$g : X \rightarrow \mathbb{R}^n.$$

In the autonomous system $\dot{x} = g(x)$, the vector $g(x)$ gives the instantaneous velocity at the state x .

Definition: Linear and nonlinear vector fields

A vector field $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is **linear** if

$$f(au + bv) = af(u) + bf(v)$$

for all $u, v \in \mathbb{R}^n$ and all $a, b \in \mathbb{R}$. Equivalently,

$$f(x) = Ax$$

for some constant $n \times n$ matrix A . A vector field that is not linear is **nonlinear**.

Question: Why is the pendulum nonlinear?

Compare

$$\ddot{\theta} + \frac{g_0}{L}\theta = 0 \quad \text{and} \quad \ddot{\theta} + \frac{g_0}{L}\sin\theta = 0, \quad g_0 > 0, \quad L > 0,$$

where g_0 is the magnitude of gravitational acceleration and L is the pendulum length.

Solution. With $\omega = \dot{\theta}$, the systems are

$$\begin{aligned} \dot{\theta} &= \omega, & \dot{\omega} &= -\frac{g_0}{L}\theta, \\ \dot{\theta} &= \omega, & \dot{\omega} &= -\frac{g_0}{L}\sin\theta. \end{aligned}$$

The first is linear. The second is nonlinear because, in general,

$$\sin(a\theta_1 + b\theta_2) \neq a\sin\theta_1 + b\sin\theta_2.$$

$$\lim_{\theta \rightarrow 0} \frac{\sin\theta}{\theta} = 1.$$

Thus $\sin\theta$ is well approximated by θ when $|\theta|$ is small. Replacing $\sin\theta$ by θ gives the linear small-angle model.

Visualization: Time graph versus phase trajectory

For a state $x = (q, v)$ with $v = \dot{q}$:

time graph	phase trajectory
$(t, q(t))$	$(q(t), v(t))$
one component versus time	the complete state in phase space

Definition: Phase portrait

A **phase portrait** is a diagram of the qualitatively distinct trajectories and equilibria in phase space, with the direction of increasing time marked on each non-equilibrium trajectory.